

TimeTensors Forecasting Benchmark

Unified task and controlled experiment design

Thesis project

Standalone method note

Abstract

This note isolates the scientific formulation from implementation and cluster operations. It is intended to be readable on its own and reusable when drafting a paper introduction or method section.

1 Forecasting population

Let $x_u(1:T)$ denote user u in a multivariate population $\mathcal{U}$. A sample anchored at t is

$$X_{u,t} = x_u(t - L + 1:t), \quad Y_{u,t} = x_u(t + 1:t + H).$$

Chronological splits own target dates; context may cross the preceding split boundary. One run seed controls population partitioning, sampling, initialization, and optimization.

2 Unified forecaster

A configured forecaster is the composition

$$\hat{Y} = g_{\text{out}}(f_{\theta}(g_{\text{in}}(X), C_{\text{past}}, C_{\text{future}})),$$

where g_{in} and g_{out} define normalization, f_{θ} is a trainable or frozen backbone, and covariates are present only for capable models. Unsupported non-empty covariates raise rather than disappearing.

3 Controlled factors

The benchmark varies one declared family at a time: constant-window policy, sampling and batch size, normalization, reference model, training loss, linear architecture, centralized versus per-user fitting, and frozen foundation model. Other training controls remain common within a comparison.

4 Population-aware evaluation

For a pointwise loss ℓ , the global estimator averages all evaluated user-window contributions. Equal-user reporting first computes $\bar{\ell}_u$ per user and then averages over users. The worst-user statistic averages the largest $\lceil 0.1|\mathcal{U}| \rceil$ values of $\bar{\ell}_u$. Seed dispersion includes every stochastic choice governed by the run seed.

5 Reproducibility boundary

Each prepared dataset variant is identified by its tensor-building configuration. Each experiment is selected through a schema-1 manifest containing scientific identity, pipeline parameters, and seeds. Reports accept only completed manifests and record their exact inputs.